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Thymus Vulgaris in Respiratory Diseases: Mechanisms and Clinical Evidence

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Abstract

Background. *Thymus vulgaris* is a medicinal plant rich in bioactive compounds, particularly thymol and carvacrol, with antimicrobial, antiviral, anti-inflammatory, immunomodulatory, antioxidant, and bronchodilatory properties relevant to respiratory diseases.

Aim. To summarize the current evidence on the mechanisms of action and therapeutic potential of *Thymus vulgaris* in respiratory diseases.

Material and methods. A narrative review of review articles, preclinical studies, and clinical trials published between 2000 and 2024 was performed using the PubMed and Google Scholar databases.

Results. Preclinical studies demonstrate that *Thymus vulgaris* exhibits antibacterial, antiviral, anti-inflammatory, immunomodulatory, antioxidant, and bronchodilatory activities through inhibition of pathogens, modulation of inflammatory pathways, reduction of oxidative stress, and relaxation of bronchial smooth muscle. Clinical studies, although limited, suggest that

thyme preparations may improve respiratory symptoms, enhance selected pulmonary function parameters, and support recovery in viral respiratory infections, including COVID-19.

Conclusions. *Thymus vulgaris* shows promise as a supportive phytotherapeutic agent in respiratory diseases. However, larger, well-designed randomized clinical trials are required to confirm its efficacy, safety, and optimal clinical use.

Keywords: *Thymus vulgaris*; essential oil; thymol; respiratory tract diseases; anti-inflammatory activity; antimicrobial activity; antiviral activity

1. Introduction

Medicinal plants have for centuries constituted an important component of both traditional and modern medicine, serving as a source of numerous biologically active compounds used in the treatment of many diseases. Among them, species belonging to the Lamiaceae family occupy a special place, as they are characterized by a high content of essential oils with a broad spectrum of pharmacological activity. One of the best-studied representatives of this family is *Thymus vulgaris*, a perennial subshrub widely cultivated in Europe. The medicinal raw material consists of the aerial parts of the plant, used in the form of dried herb as well as essential oil obtained by steam distillation. Thyme essential oil contains mainly phenolic monoterpenes, including thymol and carvacrol, which are responsible for its antibacterial, anti-inflammatory, antioxidant, and antiseptic properties. The presence of flavonoids and other terpenoids further enhances the biological activity of the plant [1,2,3].

In recent years, particular research interest has focused on the use of thyme in respiratory diseases. It has been demonstrated that extracts and essential oil from *T. vulgaris* exhibit spasmolytic and antimicrobial effects, which are of significant importance in the treatment of cough, bronchitis, and upper respiratory tract infections. The mechanism of action involves both inhibition of bacterial and viral growth, as well as reduction of inflammation and relaxation of bronchial smooth muscle, which facilitates the clearance of retained secretions. Phenolic compounds present in the essential oil also show antioxidant activity, reducing oxidative stress associated with inflammatory processes in the respiratory tract. In the context of increasing microbial resistance to antibiotics and the search for safe supportive therapies, thyme represents a promising phytotherapeutic raw material in the treatment of respiratory disorders [4,5].

2. *Thymus vulgaris* botanical description

Thymus vulgaris, commonly known as garden thyme, is a perennial, strongly aromatic plant belonging to the mint family Lamiaceae. It is a low-growing, woody, and highly branched shrub with a height of 20–40 cm. Thyme is most commonly cultivated in two- or three-year cycles, although in colder regions the plants can be damaged by winter frost. In countries such as Scandinavia, Eastern Europe, and Canada, thyme is often grown as an annual. In western and central Europe, two main forms are distinguished: *Thymus vulgaris* f. *capitatus*, the French annual variety, which is less cold-hardy, and *Thymus vulgaris* f. *vercillatus*, the German perennial variety, which is more robust and long-lived. Native to the Mediterranean region, as well as parts of North Africa and Asia, thyme is now cultivated worldwide due to its adaptability and economic importance.

The plant flourishes in dry, sunny environments and prefers light, coarse, and well-drained soils that are often unsuitable for many other crops. It forms a compact, bushy structure and produces numerous small, bilabiate flowers that may be white, yellow, or purple, typically blooming in spring. Thyme flowers secrete nectar that attracts insects, while its fruit is a four-part schizocarp containing light-brown seeds. When its leaves are gently crushed, they release a pleasant, sweet, and balsamic fragrance characteristic of aromatic herbs.

Thyme is closely related to the oregano genus *Origanum* and has been valued for centuries both as a culinary seasoning and a medicinal herb. In addition to its traditional uses, scientific research has highlighted its antioxidant and antimicrobial properties. Today, *T. vulgaris* is cultivated commercially in many countries for dried leaves, plant extracts, essential oils, and oleoresins. Thanks to its intense aroma and flavour, it is widely used in the food industry as a seasoning and natural preservative [6,7,8,9].

The genus *Thymus* is highly diverse, comprising around 220 species and demonstrating significant variation in both morphology and chemical composition, especially in the content of essential oils. While *Thymus vulgaris* (garden thyme) is the most widely known and cultivated species, other important members include *Thymus serpyllum* (wild or creeping thyme), valued for its medicinal properties and adaptability to dry, sandy, or stony soils. These species, along with a few others within the genus, are notable not only for culinary uses but also for their therapeutic potential, aromatic oils, and applications in cosmetics and the food industry.

The diversity within the genus makes thyme a rich source of both traditional and modern medicinal and culinary resources [10,11,12].

Category	Feature	Description	References
Botany	Plant type	Perennial subshrub	[3] [4] [6]
	Height	20–40 cm	
	Habit	Woody, highly branched	
	Leaves	Small, aromatic	
	Flowers	Bilabiate; white to purple	
	Flowering	Spring	
	Fruit	Four-part schizocarp	
Ecology	Climate	Warm, dry	[3] [6]
	Light	Full sun	
	Soil	Light, well-drained	
	Frost tolerance	Low to moderate	
Uses	Culinary	Spice	[3] [4] [6]
	Medicinal	Antimicrobial, antioxidant	
	Industry	Extracts, essential oils	
	Food industry	Natural preservative	

Table 1. Overview of botanical characteristics, ecology, forms, and uses of *Thymus vulgaris*

3. Bioactive compounds of *Thymus vulgaris*

Thymus vulgaris is a rich source of chemical compounds with well-documented biological activity, particularly in the form of essential oil, which constitutes a key pharmacological raw material. This oil is a complex mixture of volatile terpenoid and phenolic compounds, the composition of which varies depending on the chemotype, habitat conditions, growing season, and extraction methods [13,14,15].

3.1. Essential oil and main constituents

The essential oil of *T. vulgaris* represents the main bioactive fraction and exhibits considerable chemotypic variability. In many studies, this oil is classified as a thymol chemotype, characterized by a dominant content of thymol along with significant concentrations of p-cymene, γ -terpinene, and carvacrol [13, 14].

In a GC–MS analysis of thyme oil from Romania, thymol accounted for ~47.6%, γ -terpinene ~30.9%, and p-cymene ~8.4%, indicating a thymol chemotype for this plant material [13].

In another study on oil obtained from Slovakia, the main constituents were thymol (48.1%), p-cymene (11.7%), 1,8-cineole (6.7%), γ -terpinene (6.1%), and carvacrol (5.5%) [14].

The oil composition is generally dominated by monoterpenes and their derivatives, among which phenolic monoterpenes (e.g., thymol and carvacrol) are considered the primary contributors to antibacterial and anti-inflammatory activity [15].

3.2. Other volatile and phenolic compounds

In addition to the major terpenes, thyme essential oil contains a number of other compounds which, although present in smaller amounts, may significantly contribute to its biological properties. These include α -pinene, β -myrcene, terpineol, caryophyllene, and other terpene hydrocarbons [13,15].

Moreover, *T. vulgaris* chemotypes differ in terms of dominant compounds; besides the thymol chemotype, linalool, geraniol, and sabinene hydrate chemotypes have also been identified, reflecting the high chemical diversity of the species and the potential impact of this variability on the biological activity of the oils [15].

3.3. Non-volatile constituents and phenolic extracts

Although the essential oil is the most extensively studied raw material, *T. vulgaris* also contains other bioactive compounds beyond the volatile oil fraction. Studies on polar extracts indicate the presence of flavonoids (e.g., luteolin, apigenin, naringenin) and phenolic acids such as

rosmarinic acid and caffeic acid, which exhibit strong antioxidant activity and modulate inflammatory processes [15].

These compounds, together with the main phenolic monoterpenes, form a complex chemical profile of *T. vulgaris*, which may contribute to synergistic biological effects observed in numerous in vitro and in vivo models.

The chemical profile of *T. vulgaris* is highly complex and dependent on chemotype and environmental conditions. The essential oil containing thymol as the main component is the most frequently described bioactive fraction, alongside significant contributions of p-cymene, γ -terpinene, and other terpenes. These components, together with other phenolic and terpenoid compounds, contribute to the broad spectrum of pharmacological activity of thyme, making it an attractive raw material for the treatment of respiratory diseases and other disorders [13,14,15].

4. Antibacterial and Antifungal Activity of Thyme

Preclinical studies

Thyme essential oil, obtained from the leaves and flowering tops of *Thymus vulgaris* and other *Thymus* species from family Lamiaceae via steam distillation, is rich in bioactive compounds. Its primary constituents, thymol and carvacrol, typically make up 20–40% of the oil, accompanied by smaller amounts of borneol, cineole, linalool, menthone, p-cymene, pinene, and triterpenic acids.

The essential oil of *Thymus vulgaris* (TEO) is widely used as an adjunctive therapeutic agent in the management of various disorders, particularly upper respiratory tract infections, where it is valued for its expectorant properties. Its rich phytochemical composition accounts for its significant biological activity. Numerous studies have demonstrated that TEO exhibits antiviral, antibiofilm, antibacterial, and antifungal effects, highlighting its potential as a natural antimicrobial agent in phytotherapy and biomedical research [16,17].

The antibacterial activity of thyme is largely attributed to thymol, which disrupts bacterial cell membranes, leading to membrane depolarization and cell death as demonstrated in laboratory-based assays on bacterial cells. Thymol treatment damaged the cell membrane of *S. aureus*, with membrane damage rates increasing from 0.26% to 7.82% at 500 $\mu\text{g/mL}$. Scanning Electron Microscopy confirmed that after 6 hours, some cell membranes were severely deformed or

completely destroyed. Membrane potential assays showed significant depolarization at 200–500 µg/mL thymol, indicating that thymol induces membrane damage and depolarization, leading to bacterial cell death.

Experimental results based on in vitro cellular experiments suggest that thymol at a concentration of 300 µg/mL can cause damage to the cell membrane and decrease NADPH levels while increasing NADP⁺ and lipid peroxidation. Under these conditions, thymol has the potential to induce membrane rupture and disrupt cellular homeostasis. NADPH (reduced nicotinamide adenine dinucleotide phosphate) is a crucial coenzyme that acts as an electron donor in reductive reactions, providing the reducing power necessary for biosynthesis and maintaining cellular antioxidant defenses, thereby protecting cells from oxidative stress, DNA damage, and apoptosis. NADP⁺ is its oxidized form, and an increase in NADP⁺ indicates a disturbance in the cellular redox balance. Moreover, NOX2 (NADPH oxidase 2) plays a key role in maintaining the NADPH/NADP⁺ balance within cells. Our study confirms that NOX2 is a potential downstream target of thymol, which may provide specific guidance for its application. Collectively, these findings provide new insights into the molecular mechanisms underlying the antibacterial and antifungal actions of thymol [18].

In the study by Gislene et al., 2000, extracts from jambolan and thyme, at a concentration of 50 µg/mL, were able to inhibit the growth of *Pseudomonas aeruginosa* under laboratory conditions. Furthermore, the combination of thyme extract at 20 µg/mL with ampicillin effectively suppressed the growth of *Klebsiella pneumoniae* and *Enterobacter aerogenes*, demonstrating a synergistic antimicrobial effect [19].

Thymus vulgaris essential oil and extracts exhibit antimicrobial activity against a variety of bacteria and fungi. According to Nzeako et al. (2006), thyme oil inhibited microbial growth in vitro using microbial culture assays, whereas some organisms were not killed. The zones of inhibition produced by the oil varied among species, from 14.7 mm for *P. aeruginosa* to 33.7 mm for *S. aureus*, indicating partial growth inhibition without a bactericidal effect. Aqueous extracts showed weaker or comparable activity, with zones ranging from 10 mm for *Escherichia coli* to 34.3 mm for *S. aureus*. These results suggest that thyme oil is particularly effective against certain bacteria and fungi, likely due to bioactive constituents such as thymol and carvacrol that disrupt microbial cell membranes [20].

In the study by Thosar et al. (2013), the antimicrobial activity of *Thymus vulgaris* essential oil was evaluated against common oral pathogens in vitro using microbial culture models. The

results demonstrated that *Escherichia coli* was the most susceptible bacterium, with a minimum inhibitory concentration (MIC) of 2 µl/ml and a minimum bactericidal concentration (MBC) of 8 µl/ml. *Candida albicans* exhibited moderate sensitivity, with both MIC and minimum fungicidal concentration (MFC) at 16 µl/ml. Gram-positive bacteria showed lower susceptibility, with *Enterococcus faecalis* presenting MIC and MBC of 32 µl/ml, and *Staphylococcus aureus* MIC and MBC of 32 µl/ml. These findings indicate that thyme oil exerts strong antimicrobial effects, particularly against Gram-negative bacteria and fungi [17].

In the study by Li et al. (2022), the antibacterial effect of thymol on *Staphylococcus aureus* ATCC25923 was investigated in vitro using a bacterial culture model. Thymol at 300 µg/mL effectively suppressed bacterial growth, consistent with previous reports indicating a minimum inhibitory concentration of 0.2 mg/mL. Growth curve assays at lower concentrations (9.4–150 µg/mL) demonstrated that 150 µg/mL strongly inhibited growth, while 75 µg/mL showed a moderate inhibitory effect. Lower thymol concentrations also shortened the time for the bacteria to reach the stationary phase by 1–4 hours, indicating that thymol can inhibit *S. aureus* growth even at relatively low concentrations [18].

Clinical studies

To date, no clinical trials have been conducted in humans to directly and rigorously assess the antibacterial activity of *Thymus vulgaris* or its main constituent, thymol. Most available evidence comes from in vitro studies or animal models, which have demonstrated strong bactericidal properties and inhibition of biofilm formation. The lack of clinical data highlights the need for well-designed human studies to confirm the efficacy and safety of thyme as a potential antibacterial agent [21,22].

5. Antiviral activity of Thyme

Preclinical studies

Thymus vulgaris, particularly through its primary bioactive compound thymol, has been extensively investigated for its potential antiviral properties, demonstrating the ability to interfere with critical viral proteins, inhibit viral replication, and modulate viral transcription, thus highlighting its promise as a natural antiviral agent.

The impact of essential oils, including thymol, on their antiviral potential has been investigated. The findings suggest that these compounds may play a significant role in limiting the progression of viral infections.

The main components of selected essential oils, known for their antimicrobial properties, were subjected to molecular docking analysis against the S1 receptor-binding domain of the spike (S) glycoprotein, which is considered a key target for modern antiviral drugs. The aim was to evaluate their potential inhibitory effects based on the strength and nature of their binding interactions. In this context, particular attention was given to natural compounds present in essential oils that exhibit a broad spectrum of biological activities, including antiviral effects.

Molecular docking is a tool that enables the analysis of interactions between molecules and the prediction of their binding modes and affinities within formed complexes. A specific type of this approach is protein–ligand docking, which is widely used in modern structure-based drug design. This method is considered an effective and cost-efficient alternative to experimental techniques used to determine interactions between a protein receptor and a ligand. The analysis was performed using in silico molecular modeling approaches.

The analysis utilized the crystal structure of the SARS-CoV-2 spike (S) glycoprotein. This protein consists of two subunits: S1 and S2. The S1 subunit contains the receptor-binding domain (RBD), which is responsible for interaction with the peptidase domain of angiotensin-converting enzyme 2 (ACE2), initiating the attachment of the virus to the host cell. In contrast, the S2 subunit is involved in the fusion of viral and host cell membranes. Due to its function, the S1 subunit is of key clinical importance—its inhibition may induce conformational changes in the S protein and thereby block the early stages of viral infection.

The results indicate that selected monoterpenes, terpenoid phenols, and phenylpropanoids—such as anethole, cinnamaldehyde, carvacrol, geraniol, cinnamyl acetate, L-4-terpineol, thymol, and pulegone—exhibit significant antiviral potential. These compounds, found in essential oils derived from plants belonging to families such as Lamiaceae, Lauraceae, Myrtaceae, Apiaceae, Geraniaceae, and Fabaceae, may interact with the viral spike protein and contribute to inhibiting its activity [23,24].

Moreover, essential oils obtained from *Thymus vulgaris*, as well as from other plants including *Cymbopogon citratus* (DC.) Stapf and *Rosmarinus officinalis* L., exhibit significant antiviral activity against HIV-1 by targeting the Tat protein, which plays a key role in the viral replication

cycle and represents an important target for antiviral therapies. These findings are based on laboratory-based experimental studies.

Tat functions as a transcriptional transactivator, increasing the affinity of the P-TEFb kinase complex for the short viral RNA segment TAR (trans-activating region), which leads to enhanced synthesis of elongated viral transcripts and allows the virus to transition from the latent phase to active replication.

These essential oils are capable of inhibiting the binding of Tat to TAR RNA. The strongest effect was observed for the *R. officinalis* oil at concentrations of 0.25–0.5 µg/mL, while for *T. vulgaris* and *C. citratus* the effective ranges were 3–6 µg/mL and 6–12 µg/mL, respectively. Furthermore, *T. vulgaris* and *C. citratus* oils reduced HIV-1 transcription by 52% and 60% at applied concentrations of 0.30 ng/mL and 0.65 ng/mL, confirming their potential to inhibit viral activity [23,25].

Clinical studies

The antiviral potential of *Thymus vulgaris* has also been evaluated in clinical settings. In a randomized clinical trial, the therapeutic effect of thyme essential oil was assessed in patients diagnosed with coronavirus disease 2019 (COVID-19), a viral respiratory infection caused by SARS-CoV-2.

A total of 83 patients were randomly assigned to a control group receiving standard therapy or to an intervention group receiving thyme essential oil in addition to conventional treatment. Patients in the intervention group were instructed to consume 5 mL of thyme essential oil three times daily for seven days. Clinical symptoms and selected laboratory parameters were evaluated before and after the intervention.

After one week of treatment, patients receiving thyme demonstrated a significantly greater improvement in several clinical symptoms compared with the control group. Fever was significantly reduced in the intervention group ($p < 0.027$), while marked improvements were observed in cough, dyspnea, muscular pain, headache, anorexia, weakness and lethargy, fatigue, and chest wall pain ($p < 0.001$). Additionally, dizziness showed significant improvement ($p < 0.003$).

Laboratory analyses also revealed statistically significant changes following thyme administration. Blood urea nitrogen (BUN) levels and neutrophil counts decreased significantly in the intervention group compared with the control group ($p < 0.004$ and $p < 0.001$,

respectively). Calcium levels also showed a significant change ($p < 0.034$), while lymphocyte counts increased significantly after treatment ($p < 0.001$), suggesting a potential immunomodulatory effect [26].

Additional clinical evidence supporting the antiviral potential of *Thymus vulgaris* was reported in a cohort study involving 161 patients with PCR-confirmed COVID-19. Patients were divided into two groups: an exposed group ($n = 86$) receiving thyme extract powder (5 g dissolved in 100 mL of warm water every 8 hours) in combination with conventional therapy, and an unexposed group ($n = 75$) receiving standard treatment alone.

After three days of treatment, patients receiving thyme showed statistically significant improvements in several clinical symptoms, including cough ($p < 0.01$), dyspnea ($p < 0.001$), dizziness ($p < 0.029$), fatigue ($p < 0.001$), anorexia ($p < 0.001$), chest pain ($p < 0.001$), fever ($p < 0.024$), ageusia ($p < 0.029$), and anosmia ($p < 0.001$).

Laboratory findings also demonstrated significant immunological changes in the thyme-treated group. Neutrophil counts decreased significantly ($p < 0.01$), while lymphocyte levels increased markedly ($p < 0.001$). In addition, serum calcium levels increased significantly ($p < 0.03$), and blood urea levels showed a significant reduction ($p < 0.01$).

Importantly, viral clearance assessed by PCR testing on day 10 revealed a significantly higher proportion of negative results in the thyme-treated group compared with the control group (97.6% vs 49.3%, $p < 0.001$) [27].

Overall, these findings suggest that *Thymus vulgaris* preparations may contribute to the improvement of clinical symptoms and promote viral clearance in patients with viral respiratory infections. The observed changes in inflammatory and immune parameters further support the potential antiviral and immunomodulatory properties of thyme. Nevertheless, larger randomized clinical trials are required to confirm these findings and to better define the role of thyme-based preparations in the management of viral respiratory diseases.

6. Anti-inflammatory, immunomodulatory and bronchodilatory activity

Preclinical studies

Thymus vulgaris exhibits pronounced anti-inflammatory, antioxidant and immunomodulatory effects in the respiratory tract, which may contribute to the attenuation of airway hyperresponsiveness and inflammatory processes associated with respiratory disorders.

Airway inflammation is commonly driven by an excessive Th2-type immune response characterized by increased production of cytokines such as IL-4, IL-5, IL-9 and IL-13. These mediators stimulate IgE synthesis, bronchial epithelial proliferation, goblet cell metaplasia and excessive mucus secretion. Additionally, IL-5 and eotaxin-1 promote eosinophil recruitment to inflamed tissues, while reactive oxygen species (ROS) contribute to tissue damage and persistence of inflammation. Such immunological and structural alterations are characteristic of allergic airway diseases, particularly asthma.

The anti-inflammatory potential of thyme essential oil was investigated in an ovalbumin (OVA)-induced rabbit model of airway inflammation. Animals exposed to OVA exhibited clinical manifestations of airway dysfunction, including tachypnea, wheezing and sneezing. Histopathological examination revealed marked inflammatory infiltration, goblet cell metaplasia, increased numbers of eosinophils and apoptotic cells, enhanced collagen deposition, and elevated expression of eotaxin-1 in lung tissue. Serum levels of Th2 cytokines and IgE, as well as ROS levels in pulmonary cells, were also markedly elevated, confirming the presence of pronounced inflammatory and oxidative stress responses.

Administration of thyme essential oil (OVA + thyme group) significantly alleviated these pathological changes ($p < 0.05$). The number of inflammatory cells decreased from 34 to 14/10 HPF, goblet cells from 10 to 2/10 HPF, eosinophils from 16 to 4/10 HPF, and apoptotic cells from 18 to 3/10 HPF. The percentage area of collagen fibers decreased from 11.4% to 4.2%, while immunoexpression of eotaxin-1 was reduced from strong to weak.

Similarly, serum levels of TSLP, IL-4, IL-5, IL-9, IL-13, IgE and eotaxin-1 decreased from 360, 270, 130, 85, 365, 398 and 110 pg/ml to 162, 58, 20, 42, 145, 112 and 52 pg/ml, respectively. ROS levels in lung cells were also significantly reduced from 14433.3 to 7626.5 MFI.

These effects are attributed primarily to phenolic constituents of thyme essential oil, including thymol, carvacrol, p-cymene and linalool, which exhibit anti-inflammatory, antioxidant, antiapoptotic and immunomodulatory properties. Mechanistically, these compounds inhibit the

expression of pro-inflammatory mediators such as IL-4, IL-5, IL-13 and TNF- α and suppress activation of the NF- κ B signaling pathway, thereby reducing inflammatory mediator production, inflammatory cell infiltration and airway remodeling.

Collectively, these findings suggest that *Thymus vulgaris* may attenuate airway inflammation, oxidative stress and bronchial hyperresponsiveness, indicating its potential as a supportive therapeutic agent in inflammatory respiratory disorders [28].

In addition to its anti-inflammatory and immunomodulatory effects, *Thymus vulgaris* has also demonstrated direct bronchodilatory activity in experimental models. In an in vitro study on isolated guinea-pig tracheal chains, both aqueous and macerated extracts of thyme induced significant, concentration-dependent relaxation of airway smooth muscle compared with saline ($p < 0.05$ to $p < 0.001$). The observed effects were comparable to those of theophylline at higher concentrations. A significant positive correlation between extract concentration and relaxant effect was also reported ($p < 0.01$ to $p < 0.001$). Notably, the bronchodilatory effect was markedly reduced in tissues pretreated with propranolol and chlorphenamine, suggesting the involvement of β -adrenergic receptor stimulation and histamine H1 receptor blockade. Additional mechanisms, including calcium channel inhibition and potassium channel activation, have also been proposed. These findings indicate that *Thymus vulgaris* may directly reduce bronchial smooth muscle tone, complementing its anti-inflammatory effects and supporting its potential role in the management of obstructive airway disorders [29].

Clinical studies

The potential benefits of *Thymus vulgaris* in respiratory conditions have also been investigated in clinical settings. In a randomized, triple-blind, placebo-controlled trial, the efficacy of a syrup containing thyme extract was evaluated as an adjunct therapy in children aged 5–12 years with mild to moderate exacerbations of airway inflammation.

A total of 60 patients were included in the study. In addition to standard anti-asthmatic therapy, participants received thyme syrup at a dose of 20 mg/kg body weight every 8 hours for seven days or placebo.

After the intervention, a statistically significant improvement in clinical symptoms was observed in the thyme-treated group. Persistent cough was reported in 16.7% of patients receiving thyme compared with 40% in the placebo group ($p = 0.042$). The frequency of

dyspnea decreased in both groups; however, the difference between groups was not statistically significant (16.7% vs 23.3%; $p = 0.37$). Likewise, no statistically significant difference was observed in the incidence of wheezing after treatment ($p = 0.266$).

Spirometric analysis demonstrated a significant improvement in lung function parameters in the thyme group. Post-treatment FEV₁ values were higher in the thyme group (1.98 ± 0.38 L) compared with the placebo group (1.77 ± 0.45 L; $p = 0.048$). No statistically significant differences were observed between groups in the FEV₁/FVC ratio, peak expiratory flow (PEF), or parameters reflecting airflow in the small airways.

The preparation was well tolerated and no serious adverse events were reported during the treatment period.

Overall, these findings suggest that *Thymus vulgaris* extract may serve as a beneficial adjunct therapy in respiratory disorders characterized by airway inflammation, particularly by reducing cough and improving selected pulmonary function parameters. However, the relatively small sample size and short duration of treatment indicate that further clinical studies are required to confirm its long-term efficacy and safety [30].

Additional clinical evidence supporting the use of thyme-containing preparations in respiratory disorders was demonstrated in adults with acute bronchitis and productive cough. In a prospective, double-blind, placebo-controlled multicentre trial involving 361 outpatients, treatment with a fixed combination of dry extracts of thyme herb (*Thymus vulgaris*) and primrose root resulted in a significantly greater reduction in coughing fits compared with placebo (67.1% vs 51.3%; $p < 0.0001$). Moreover, improvement in Bronchitis Severity Score (BSS) was significantly faster in the treatment group, and the preparation was well tolerated with no serious adverse events reported [31].

Similarly, in a randomized, double-blind, placebo-controlled study including 150 patients with acute bronchitis, treatment with a fixed combination of thyme and primrose root significantly reduced bronchitis severity compared with placebo, as reflected by a greater decrease in BSS scores ($p \leq 0.001$). In addition, significantly more patients in the treatment group became symptom-free by the end of the study, while the herbal preparation demonstrated very good tolerability and no serious adverse effects [32].

Activity	Main compounds / fractions	Mechanisms of action	References
Antibacterial / Antifungal	Thymol, carvacrol	Disruption of microbial cell membranes, membrane depolarization, increased permeability, oxidative imbalance ($\downarrow$ NADPH / $\uparrow$ NADP ⁺), lipid peroxidation, possible NOX2 involvement $\rightarrow$ cell death	[12] [14]
Antiviral	Thymol and other essential oil constituents	Molecular docking: interaction with SARS-CoV-2 spike (S1 RBD); inhibition of HIV-1 Tat–TAR interaction; reduction of viral transcription and replication potential	[15] [16] [17]
Anti-inflammatory and Immunomodulatory	Thymol, carvacrol, p-cymene, linalool; flavonoids	Reduction of airway inflammation in OVA model: $\downarrow$ IL-4, IL-5, IL-9, IL-13; $\downarrow$ IgE; $\downarrow$ eotaxin-1; $\downarrow$ eosinophils, goblet cells, collagen deposition; suppression of NF- κ B signaling; $\downarrow$ ROS	[20]
Bronchodilatory (spasmolytic)	Thyme extracts (composition not fully isolated)	Relaxation of bronchial smooth muscle; likely involving β -adrenergic pathways, H1 receptor blockade and Ca ²⁺ channel modulation	[21]

Table 2. Mechanisms of action of *Thymus vulgaris* in respiratory disorders

Conclusions

Thymus vulgaris is a well-characterized medicinal plant with a rich profile of bioactive compounds, particularly thymol and carvacrol, which are responsible for its broad pharmacological activity. The available preclinical evidence consistently demonstrates its antibacterial, antifungal, antiviral, anti-inflammatory, immunomodulatory, and bronchodilatory properties, mainly through membrane disruption of pathogens, modulation of inflammatory pathways, and reduction of oxidative stress. Clinical studies, although still limited in number and scale, suggest potential benefits of thyme preparations in improving respiratory symptoms and supporting recovery in airway-related disorders, including asthma exacerbations and viral infections. Overall, *T. vulgaris* shows promising potential as a supportive phytotherapeutic agent in respiratory medicine; however, further large-scale, well-designed clinical trials are necessary to confirm its efficacy, safety, and optimal therapeutic applications [4,33].

Disclosure

Author Contributions:

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